Waveform Coding

Prediction Filter:

Linear Prediction:

Prediction is used for estimating the future samples of a signal from present and past input sample values.

The prediction is said to be linear if the future sample value is a linear combination of present and past input sample values.

The predicted sample value is given by

$$\hat{x}(n) = \sum_{k=1}^{M} w_k x(n-k) \longrightarrow (1)$$

where $\hat{x}(n)$ is predicted future sample

$x(n-k) \rightarrow$ past sample

$\hookrightarrow x(n-1), x(n-2), \ldots x(n-k)$

$w_k \rightarrow$ set of multiplier co.eff.

Prediction error:

The error that happens because of prediction is called prediction error. Prediction error is the difference blt the actual sample value and the predicted sample value.

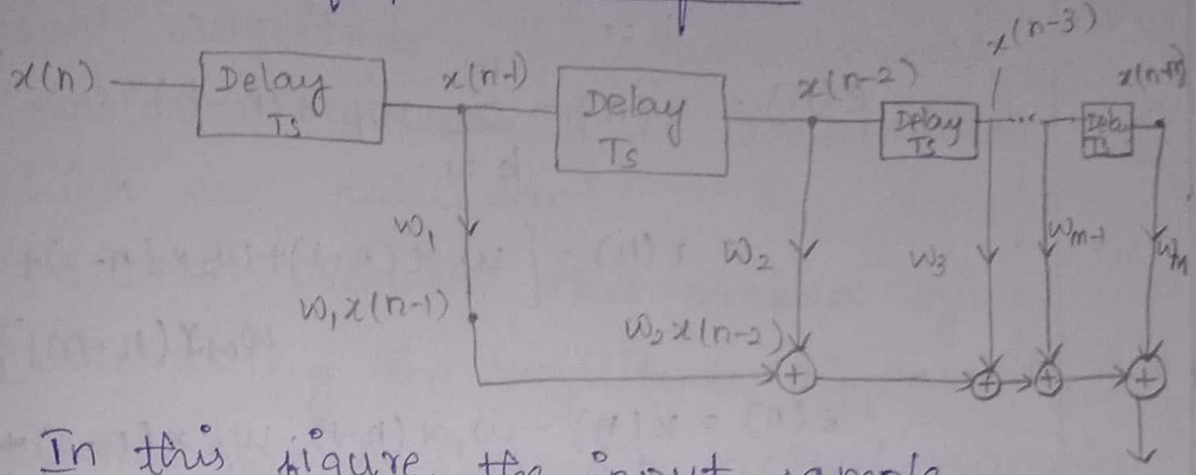
Prediction filter:

→ is a digital filter used in LPC to estimate current speech sample from previous speech samples.

Implementation of linear predictor equation

$\hat{x}(n) = \sum_{k=1}^{M} w_k x(n-k)$ is known as prediction filter.

Structure of prediction filter:



In this figure, the input sample $x(n)$ is delayed by one sampling duration T_s. Hence we get $x(n-1)$ which is the previous input sample. similarly, If $x(n-1)$ is delayed by T_s, we get $x(n-2)$ and so on. The past input samples are multiplied by multiplier co-efficients $w_1, w_2, \ldots, w_m$ and are added to get $\hat{x}(n)$. This structure is called a filter because eqn ① represents linear convolution and such convolution represents filtering. In this structure $x(n-1), x(n-2), \ldots, x(n-m)$ are the past inputs which are used for filtering with the help of multiplier co-efficients $w_1, w_2, \ldots, w_m$. This filter is of the type FIR digital filter. The filtering operation can be modified by changing the filtering coefficients

finite FIR $\leftarrow \sum_{k=0}^{m}$ not $\sum_{k=0}^{\infty}$ IIR

structure of prediction error:

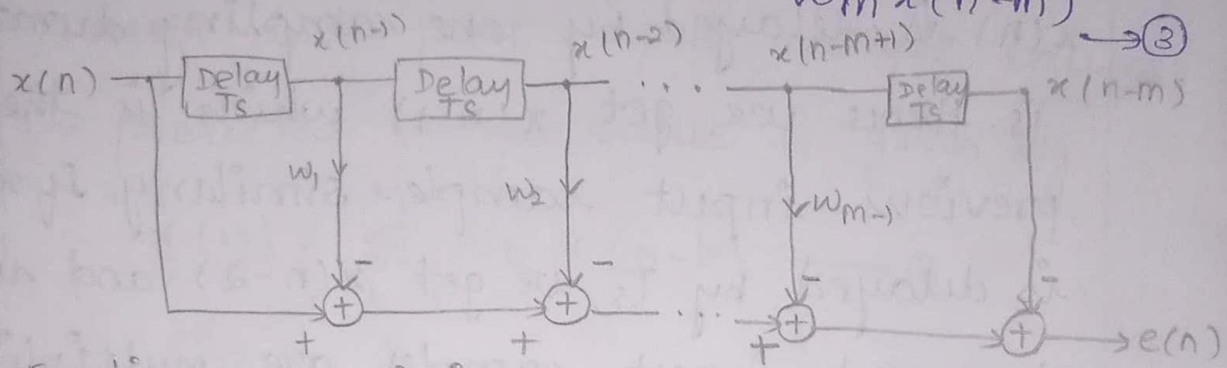
The prediction error $e(n) = x(n) - \hat{x}(n)$ → (2)

where $\hat{x}(n) = \sum_{k=1}^{m} w_k x(n-k)$

$$e(n) = x(n) - \sum_{k=1}^{m} w_k x(n-k)$$

$$= x(n) - [w_1 x(n-1) + w_2 x(n-2) + \ldots + w_M x(n-m)]$$

$$e(n) = x(n) - w_1 x(n-1) - w_2 x(n-2) + \ldots - w_m x(n-m)$$ → (3)



For linear predictive coding (write up to this).

Application of prediction filter:

⇒ Prediction filter is used in DPCM system to minimize the error applied to the DPCM quantizer.

⇒ Data compression - reduces amt of data req. to represent a speech signal.

Adv

⇒ Low bit rate.

⇒ reduces redundancy

⇒ Efficient transmission

⇒ speech compression - reduces amt of data req. to transmit speech.

DPCM - Differential Pulse Code Modulation.

Need for DPCM [or Limitation of PCM]:

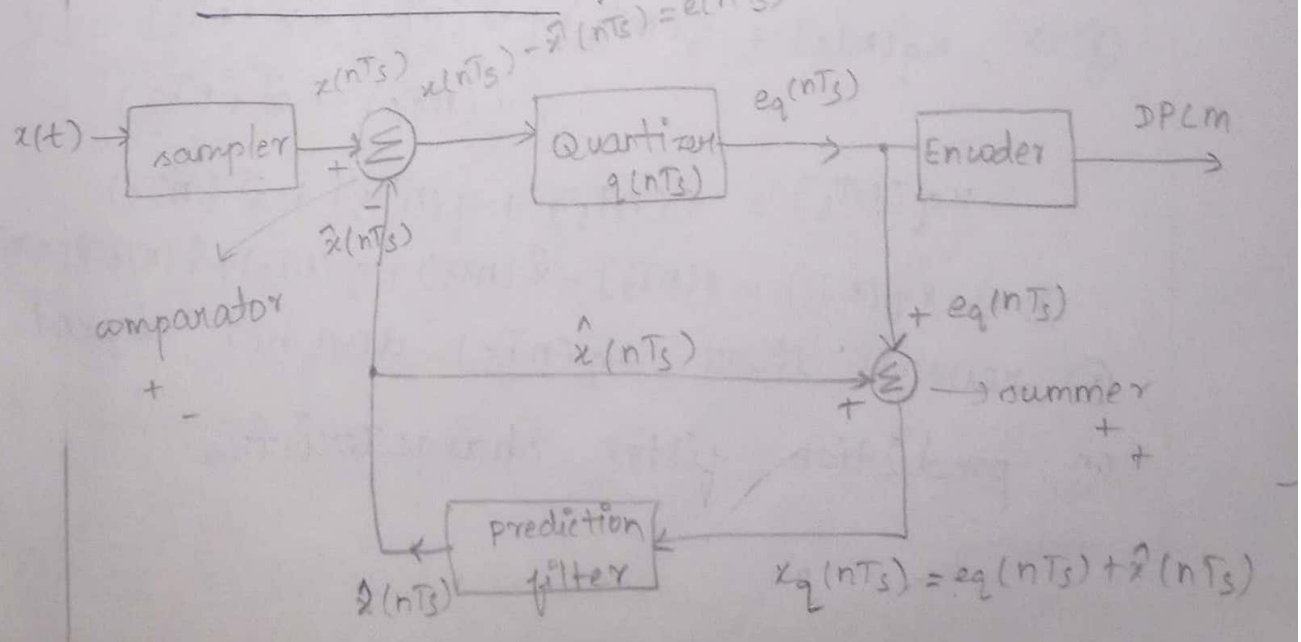
1. In PCM, the adjacent samples are of almost equal amplitude. This causes several identical PCM codes to be transmitted which are redundant.

2. DPCM overcomes this problem by transmitting the amplitude difference of two successive samples instead of actual samples. Hence fewer bits are required for DPCM.

Principle of DPCM:

In DPCM, redundancy and overall bit rate will decrease. No. of bits required to transmit one sample will also be reduced by transmitting only the difference between consecutive samples.

DPCM - transmitter:



The input signal $x(t)$ is given to the sampler. The output of sampler $x(nT_s)$ and a predicted sample $\hat{x}(nT_s)$ is fed to comparator. The output of comparator $e(nT_s)$ is sent to the quantiser where quantisation error gets added. The output of quantiser $e_q(nT_s)$ is fed to the encoder. The output of quantiser and the prediction filter's output are given to a summer and its output $x_q(nT_s)$ is passed to the prediction filter.

$$x_q(nT_s) = e_q(nT_s) + \hat{x}(nT_s) \longrightarrow (1)$$

$$e_q(nT_s) = e(nT_s) + q(nT_s) \longrightarrow (2)$$

where $q(nT_s)$ is the quantization error.

$$(1) \Rightarrow x_q(nT_s) = e(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$e(nT_s) = x(nT_s) - \hat{x}(nT_s)$$

$$x_q(nT_s) = x(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s) \longrightarrow (5)$$

$$x(nT_s) = e(nT_s) + \hat{x}(nT_s) \longrightarrow (4) \quad \text{from sampler}$$

sub (4) in (3)

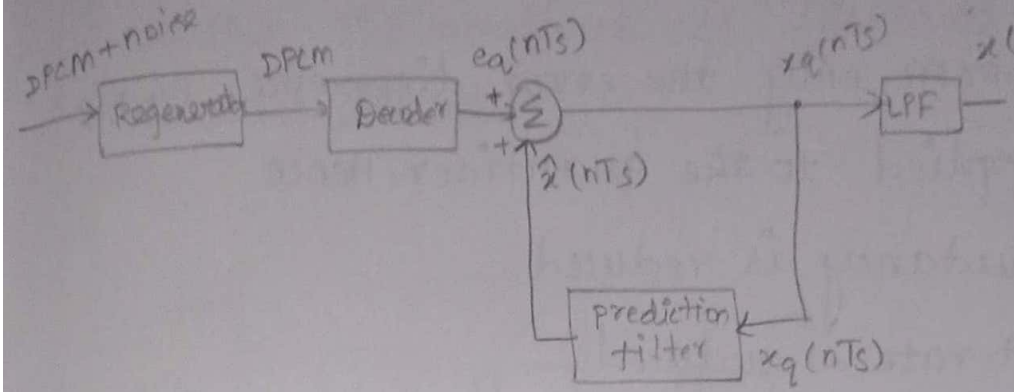
$$(3) \Rightarrow x_q(nT_s) = e(nT_s) + \hat{x}(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$x_q(nT_s) = e(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$(3) \Rightarrow x_q(nT_s) = x(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s) = x(nT_s) + q(nT_s) \longrightarrow (5)$$

(5) reveals that $x_q(nT_s)$ does not depend on prediction filter characteristics

DPCM Receiver:



=> DPCM Rx comprises of regenerator, decoder, summer, prediction filter, LPF blocks.

=> Regenerator block receives the DPCM signal & noise signal that fed into decoder.

=> in decoder, it decodes the signal and produces the signal $e_q(nT_s)$.

=> In summer, it receives two signals one is $e_q(nT_s)$ and o/p of prediction filter $\hat{x}(nT_s)$

=> Then the o/p of summer gives $x_q(nT_s)$ that move to the LPF which is also a smoothening filter and smoothens the signal and give $x(nT_s)$ i.e msg signal.

$s)+q(nT_s)$

→ (5)

redundant information.

In PCM

If three successive samples have same amplitude, it is sent as it is

In DPCM

redundancy ↓

diff. of amplit is sent.

redundant - nearly same ... prevailed.

Advantages of DPCM over PCM:

⇒ In DPCM, only the error difference $e(nT_s)$ is applied to the quantiser. Hence

1. reducdancy is reduced.
2. Bit rate is reduced.
3. Transmission BW is reduced.

Disadvantages:

1. In DPCM, the system complexity is increased because of the use of prediction filter.

Delta Modulation:

In DPCM → $x(nT_s) - \hat{x}(nT_s) = e(nT_s)$ → encoded → T_x.

In DM → $x(nT_s) - \hat{x}(nT_s)$ → $u(n-1)T_s$

+ve = ↑ Step size $+\delta \rightarrow 1$

-ve = $-\delta \rightarrow 0$

error — encoding.

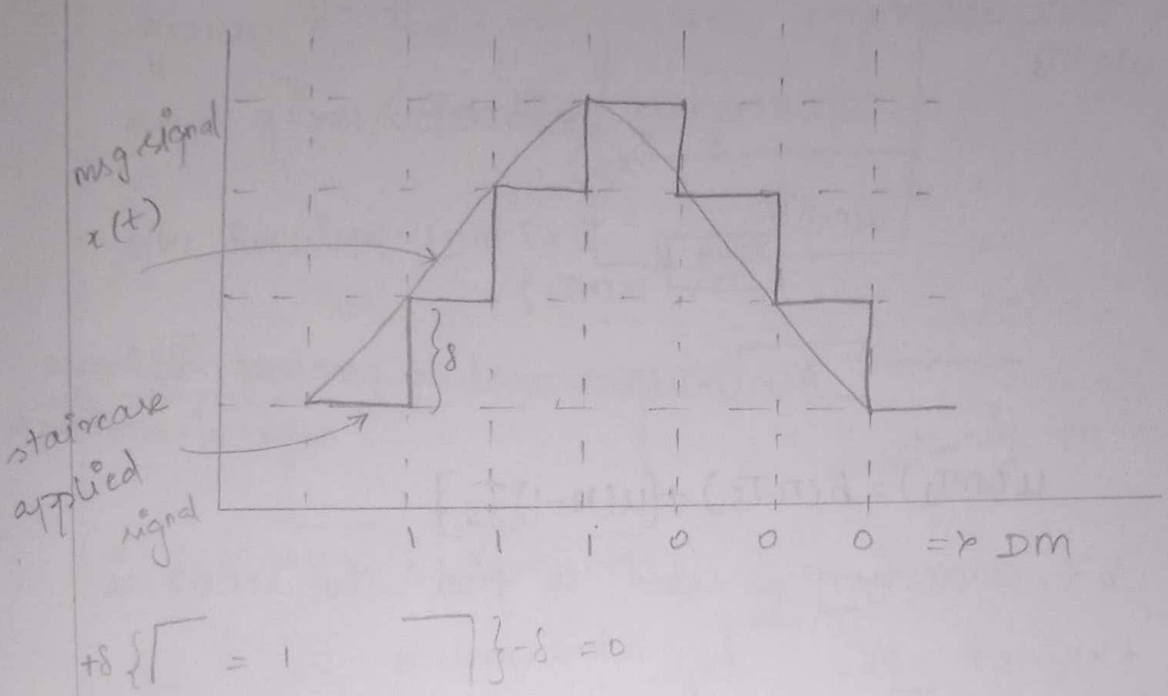
Comparison of PCM and DM:

1. PCM transmitts all bits which are used to code the sample. Hence, the signaling rate and transmission BW are large in PCM. To overcome this problem DM is used which transmitts only one bit per sample.

Operating Principle of DM:

DM transmitts only one bit per sample i.e the present sample value is compared

with previous sample value and the indication whether the amplitude is increased or decreased is sent.

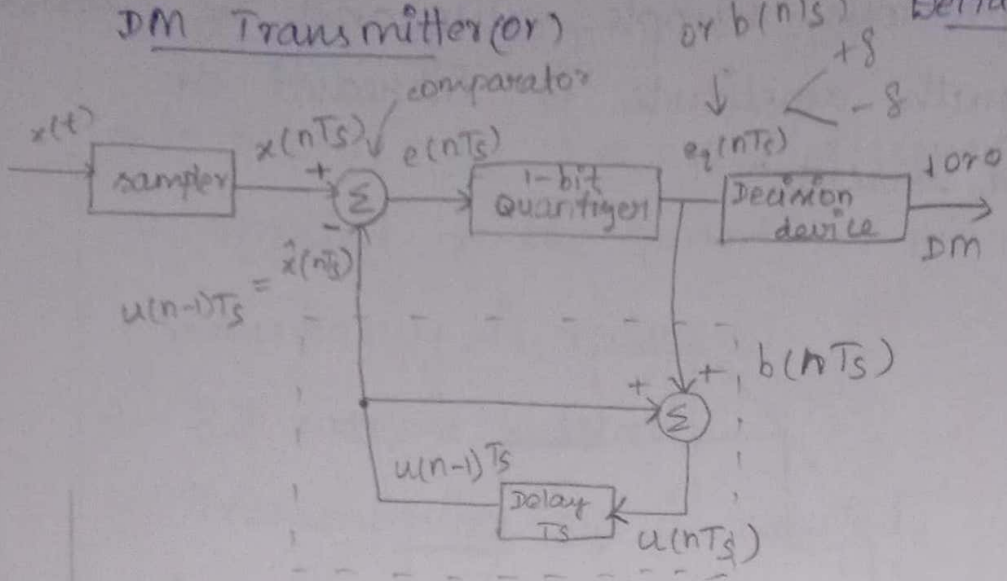


The input signal $x(t)$ is approximated to step signal (staircase approximated signal) by delta modulator. The step size is fixed. The difference between the input signal $x(t)$ and the staircase approximated signal is compared to two levels. ($+\delta$ and $-\delta$)

If the difference is positive, then the approximated signal is increased by one step size ($+\delta$) and bit '1' is transmitted.

If the difference is negative, the the approximated signal is decreased by one step size ($-\delta$) and bit '0' is transmitted.

DM Transmitter (or) Delta Modulator.



Accumulator - used to produce staircase approximated signal.

$$u(nT_s) = b(nT_s) + [u(n-1)T_s]$$

1 bit - Quantizer is used to find the error is +ve or -ve.

Accumulator consists of Delay, summer.

=> The first block of Delta modulator is sampler which is used to convert continuous signal $x(t)$ to discrete signal $x(nT_s)$.

=> Then $x(nT_s)$ is sent to comparator, and $\hat{x}(nT_s)$ is also sent to comparator and produces the output $e(nT_s)$ i.e, error signal.

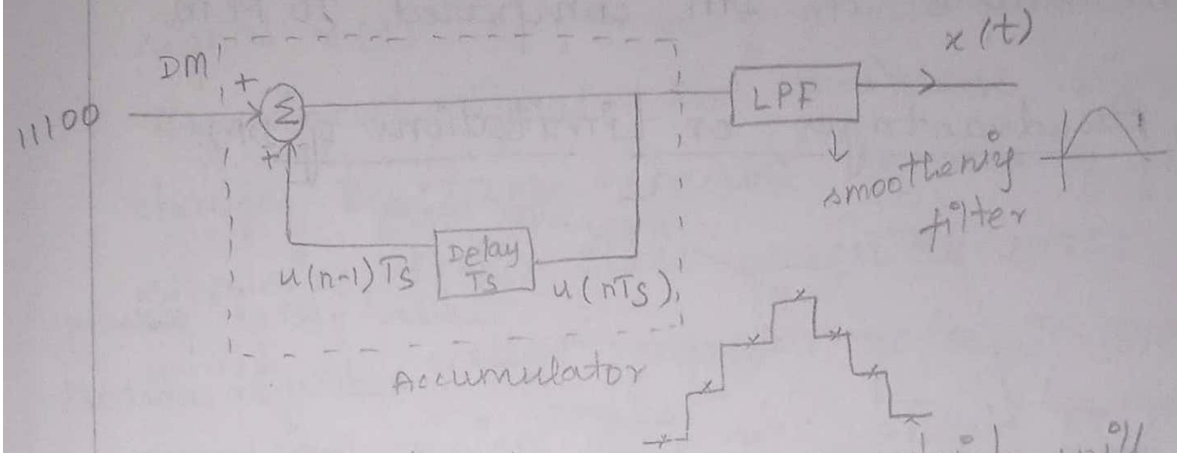
=> $e(nT_s)$ is given to 1-bit quantiser and gives $e_q(nT_s)$ or $b(nT_s)$.

=> $b(nT_s)$ is generated based on the o/p of comparator $e(nT_s)$; if $e(nT_s)$ is +ve then o/p of Quantizer $+\delta$ or else produces $-\delta$.

=> This will fed to decision device whether it transmits 1 or 0 based on o/p of

quantiser. In feedback path, there is an accumulator which consists of summer, delay unit. With the help of accumulator, we generate the staircase approximated signal and given to the comparator.

DM Receiver: (DM Rx)



Use regenerator before summer, which will remove noise.

At o/p of DM Rx, 100% reconstruction does not happen due to "quantiser".

⇒ The DM signal and the delayed signal $u(n-1)T_s$ are fed as inputs to a summer.

⇒ The o/p of summer is given to a LPF which is actually a smoothening filter.

⇒ At the output of LPF, we get back the signal $x(t)$.

⇒ The summer and the delay unit altogether acts as an accumulator.

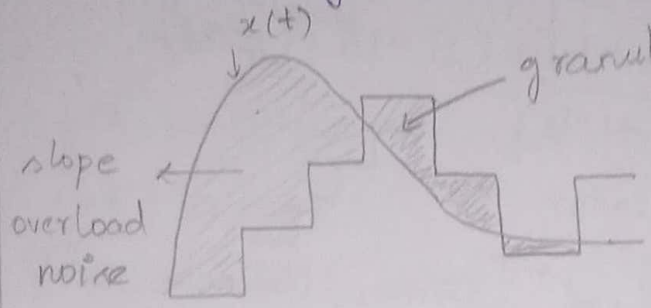
bit rate - no. of bits transmitted per sample

Advantages of DM over PCM:

=> The bit rate and transmission BW is reduced in DM as it transmits only 1 bit per sample.

=> The transmitter and receiver implementation is simple in DM compared to PCM.

Disadvantages or Limitations of DM:



when x(t) is varying slowly or constant.

=> δ is fixed.

=> To overcome, δ → ↓

when x(t) is steep

=> δ is fixed.

=> To overcome, δ → ↑

Slope overload distortion (or) noise:

This distortion arises because of large dynamic range of the input signal. Referring the diagram shown, the rate of rise of input signal x(t) is so high that the staircase signal cannot approximate since the step size δ becomes ~~too~~ small for the staircase signal to follow the steep segment of x(t). Thus there exists an error or distortion which is called as

slope overload distortion. To reduce this error, step size should be ↑ when slope of the signal $x(t)$ is high.

Granular noise:

It occurs when the step size is too large compared to small variation in the signal $x(t)$ i.e, for very small variations in the input signal, the staircase signal changes by large amount because of fixed step size. Hence there occurs an error which is called granular noise. To avoid granular noise, step size should be made small.

1. Consider a sine wave of freq. f_m and amplitude A_m is applied to a Delta modulator with step size δ. Show that slope overload distortion will occur if $A_m > \frac{\delta}{2\pi f_m T_s}$ where T_s is the sampling period of DM.

soln:

Let us assume a sinusoidal wave of freq. f_m and amplitude A_m.

$$x(t) = A_m \sin(2\pi f_m t)$$

slope of sine wave $x(t) = \left| \frac{dx(t)}{dt} \right|$

slope of DM $= \frac{\text{step size}}{\text{sampling period}} = \frac{\delta}{T_s}$

slope overload noise will occur if

slope of $x(t)$ > slope of DM

$$\left|\frac{dx(t)}{dt}\right| > \frac{\delta}{T_s}$$

$$\left|\frac{d(A_m \sin(2\pi f_m t))}{dt}\right| > \frac{\delta}{T_s}$$

$$|A_m \cos(2\pi f_m t)\, 2\pi f_m| > \frac{\delta}{T_s}$$

$$2\pi A_m f_m \,|\cos(2\pi f_m t)| > \frac{\delta}{T_s}$$

$\cos\theta_{max} = 1$

$$2\pi A_m f_m \,(1) > \frac{\delta}{T_s}$$

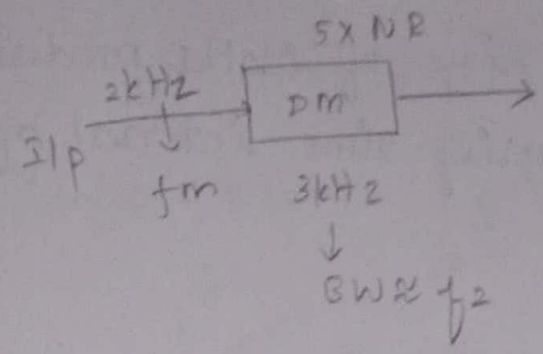
$$A_m > \frac{\delta}{2\pi f_m T_s}$$

To overcome slope overload noise,

$$A_m < \frac{\delta}{2\pi f_m T_s}$$

2. A delta modulator is designed to operate at 5 times Nyquist rate for a signal with 3kHz bandwidth. Determine the maximum amplitude of a 2kHz input sinusoid for which the delta modulator does not have slope overload. Quantisation step size is 200 mV. Derive the formula that you use.

soln:



For DM,

$f_s = 5 \times$ Nyquist rate

$= 5 \times 2 f_2$

$f_s = 10 f_2 = 10 \times 3\,kHz$

$f_s = 30\,kHz$

$T_s = \frac{1}{30k} = 0.0333\,ms$

$T_s = 33.3\,\mu s$

slope overload will not occur if

$$A_m < \frac{\delta}{2\pi f_m T_s} \longrightarrow \left|\frac{dx(t)}{dt}\right| < \frac{\delta}{T_s}$$

$$< \frac{200 \times 10^{-3}}{2\pi \times 2 \times 10^3 \times 0.0333 \times 10^{-3}}$$

$$< 478.185 \times 10^{-3}$$

$A_m < 0.478\ V$

$A_m < 478\ mV$

(*) Signal to Quantisation Noise Ratio of DM:

Assume that there is no slope overload noise. $A_m < \frac{\delta}{2\pi f_m T_s}$ ——→ (1)

where A_m → Amplitude of the input (msg) signal

f_m → max. freq. of input signal.

$\delta \rightarrow$ step size of Delta modulator

$T_s \rightarrow$ Sampling time interval of Delta modulator

Signal power:

$$P = \frac{V^2_{signal}}{R} \longrightarrow (2)$$

where $V^2_{signal} \rightarrow$ mean square value of i/p signal

Assume input as sinusoidal signal which is normalized to 1.

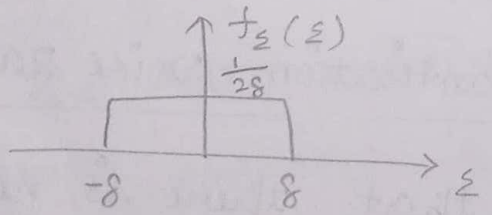
$$P = \left(\frac{Am}{\sqrt{2}}\right)^2 \longrightarrow (3)$$

Sub (1) in (3).

$$P = \frac{\left(\frac{\delta}{2\pi f_m T_s}\right)^2}{2} = \frac{\delta^2}{8\pi^2 f_m^2 T_s^2} \longrightarrow (4)$$

Noise power:

Maximum quantisation error in DM is equal to the step size δ. Let the quantisation error, we uniformly distributed over the interval $(-\delta, +\delta)$.



$$\frac{1}{b-a} = \frac{1}{\delta-(-\delta)} = \frac{1}{2\delta}$$

$$f_\varepsilon(\varepsilon) = \begin{cases} \frac{1}{2\delta} ; & -\delta < \varepsilon < \delta \\ 0 ; & \text{otherwise} \end{cases}$$

$$\text{Noise power} = \frac{V^2_{noise}}{R} \longrightarrow (5)$$

where V^2_{noise} is mean square value of noise.

$$V^2_{noise} = E(x^2) = \int_{-\infty}^{\infty} \varepsilon^2 f_\varepsilon(\delta)\, d\varepsilon$$

$$= \int_{-\delta}^{\delta} \varepsilon^2 \frac{1}{2\delta}\, d\varepsilon$$

$$= \frac{1}{2\delta} \int_{-\delta}^{\delta} \varepsilon^2\, d\varepsilon$$

$$= \frac{1}{2\delta} \left[\frac{\varepsilon^3}{3} \right]_{-\delta}^{\delta}$$

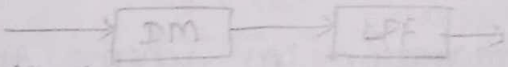
$$= \frac{1}{2\delta} \left[\frac{\delta^3}{3} + \frac{\delta^3}{3} \right]$$

$$= \frac{2\delta\delta^2}{2\delta(3)}$$

$$V^2_{noise} = \frac{\delta^2}{3} \rightarrow (6)$$

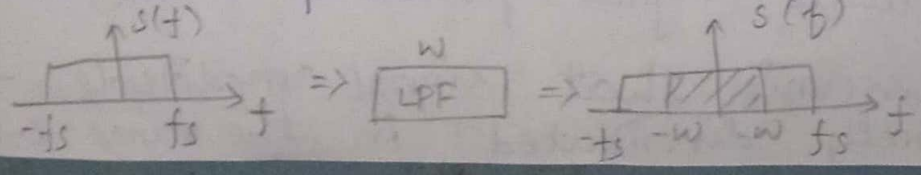
With R = 1 (normalised),

(5) => Noise power = $V^2_{noise} = \frac{\delta^2}{3} \rightarrow (7)$



Assume that

The noise power is uniformly distributed over $-f_s$ to $+f_s$ range. At the output of DM receiver, there is a low pass reconstruction filter whose cut-off freq. is 'w'. This cutoff freq. is equal to highest signal freq. The reconstruction filter passes part of the noise power at the output.



Output noise power $= \frac{w}{f_s}$ noise power $= \frac{w}{f_s} \times \frac{\delta^2}{3}$

SNR (post-filtered):

$$SNR = \frac{\text{signal power}}{\text{Noise power (post filtered)}}$$

$$= \frac{\frac{\delta^2}{8\pi^2 f_m^2 T_s^2}}{\frac{w}{f_s} \times \frac{\delta^2}{3}}$$

$$= \frac{3 f_s}{8\pi^2 f_m^2 T_s^2 w}$$

$$\boxed{SNR = \frac{3}{8\pi^2 f_m^2 T_s^3 w}} \; ; \; \left[f_s = \frac{1}{T_s}\right]$$

[Considering only the granular noise and post filtered].

1. In a single integration DM scheme, the voice signal is sampled at the rate of 64 kHz. The maximum signal amplitude is 1 V and signal BW is 3.5 kHz. i) Determine the minimum value of step size to avoid slope-overload.
ii) Determine the granular noise N_o
iii) Assume the signal to be sinusoidal and calculate signal power and SNR.
iv) Assuming that signal amplitude is uniformly distributed in the range (-1, 1).

Determine signal power and amplitude.

soln:

Given,

i) $A_m = 1V$, $f_m = f_2 = BW = 3.5kHz$, $f_s = 64kHz$

$$T_s = \frac{1}{f_s} = 0.0156\,ms = 15.6\,\mu s$$

$$A_m \leq \frac{\delta}{2\pi f_m T_s}$$

$$A_m \cdot 2\pi f_m T_s \leq \delta$$

$$1 \times 2 \times 3.14 \times 3.5 \times 10^3 \times 15.6 \times 10^{-6} \leq \delta$$

$$0.342 \leq \delta$$

$$\delta \geq 0.342$$

$\delta = 0.34V$ ✓ or $\delta = 342\,mV$.

ii) Granular noise power: N_o

$W = f_m$

$$N_o = \frac{W}{f_s} \times \frac{\delta^2}{3}$$

$$= \frac{3.5 \times 10^3}{64 \times 10^3} \times \frac{(0.34)^2}{3}$$

$$N_o = 2.107 \times 10^{-3}$$

$$N_o = 2.107\,mW$$

iii) $$P_{signal} = \frac{\delta^2}{8\pi^2 f_m^2 T_s^2} = \frac{(0.34)^2}{8\pi^2 (3.5 \times 10^3)^2 (15.6 \times 10^{-6})^2}$$

$$= 0.4916 \simeq 0.5\,W$$ ✓

$$P_{signal} = \left(\frac{A_m}{\sqrt{2}}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} = 0.5W$$ ✓

$$SNR = \frac{P_{sig}}{P_{noise}} = \frac{3}{8\pi^2 f_m^2 T_s^3 W}$$ ***

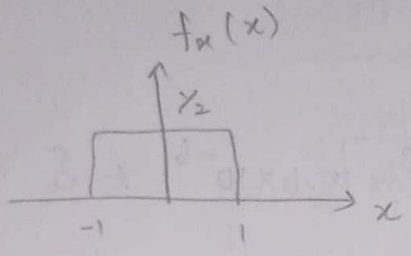
$$SNR = \frac{3}{8\pi^2 \times (3.5\times10^{3})^2 \times (15.6\times10^{-6})^3 \times 3.5\times10^{3}}$$

$$= 233.665$$

$SNR \simeq 234 \rightarrow$ This is post-filtered SNR.

$$SNR = \frac{P_{sig}}{P_{noise}} = \frac{0.5}{0.002107} = 237.3 \simeq 237.$$ ✓

iv)



$$f_x(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$P = \frac{V^2_{signal}}{R} \quad [R=1]$$

$$P = V^2_{signal} = E[X^2] = \int_{-1}^{1} x^2 f_x(x)\,dx$$

$$= \int_{-1}^{1} x^2 \frac{1}{2}\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_{-1}^{1}$$

$$= \frac{1}{2}\left[\frac{1}{3} + \frac{1}{3}\right] = \frac{1}{2}\left[\frac{2}{3}\right] = \frac{1}{3}$$

$P_{sig} = 0.3333\,W$ ✓

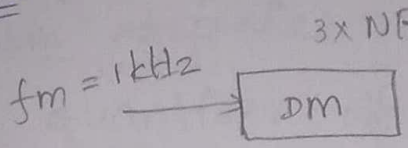
$A_{sig} = \frac{1}{2}V$ ✓

$$SNR = \frac{P_{sig}}{P_{noise}} = \frac{0.3333}{0.002107} = 158.18$$

2. A DM system is designed to operate at three times the Nyquist rate for a signal

with 3kHz BW. The quantisation step size is 250mV.
i) Determine maximum amplitude of 1kHz input sinusoid for which DM doesn't show slope overload.
ii) Determine the post-filtered output SNR for the signal of part i).

soln:



$BW \approx f_2 = 3kHz$

i) $f_s = 3 \times NR = 3 \times 2f_2 = 6f_2 = 6 \times 3kHz = 18kHz$

$T_s = \frac{1}{f_s} = 0.0556ms = 55.6\mu s$

$A_m < \frac{\delta}{2\pi f_m T_s}$

$< \frac{250 \times 10^{-3}}{2\pi \times 1 \times 10^{3} \times 55.6 \times 10^{-6}}$

$A_m < 0.7159$

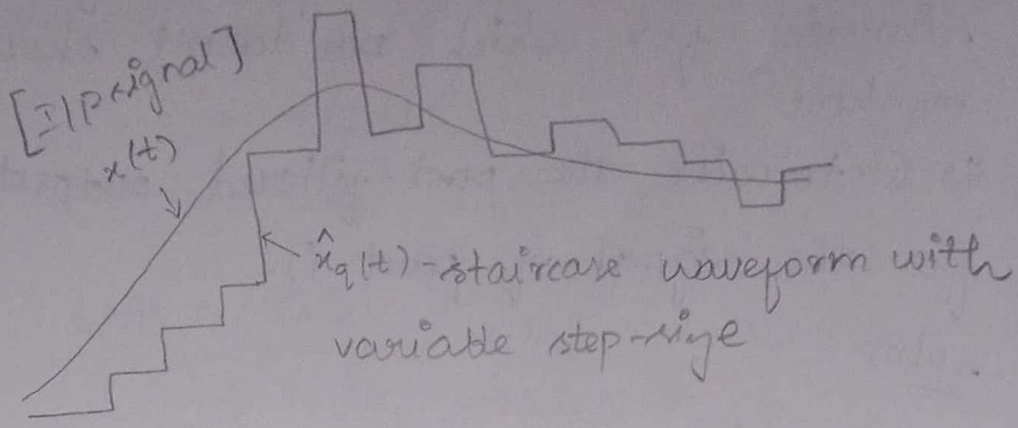
$\therefore$ max. amplitude $= 0.7\ V$

ii) $SNR = \frac{3}{8\pi^2 f_m^2 T_s^3 W}$ Here $W = 3kHz$

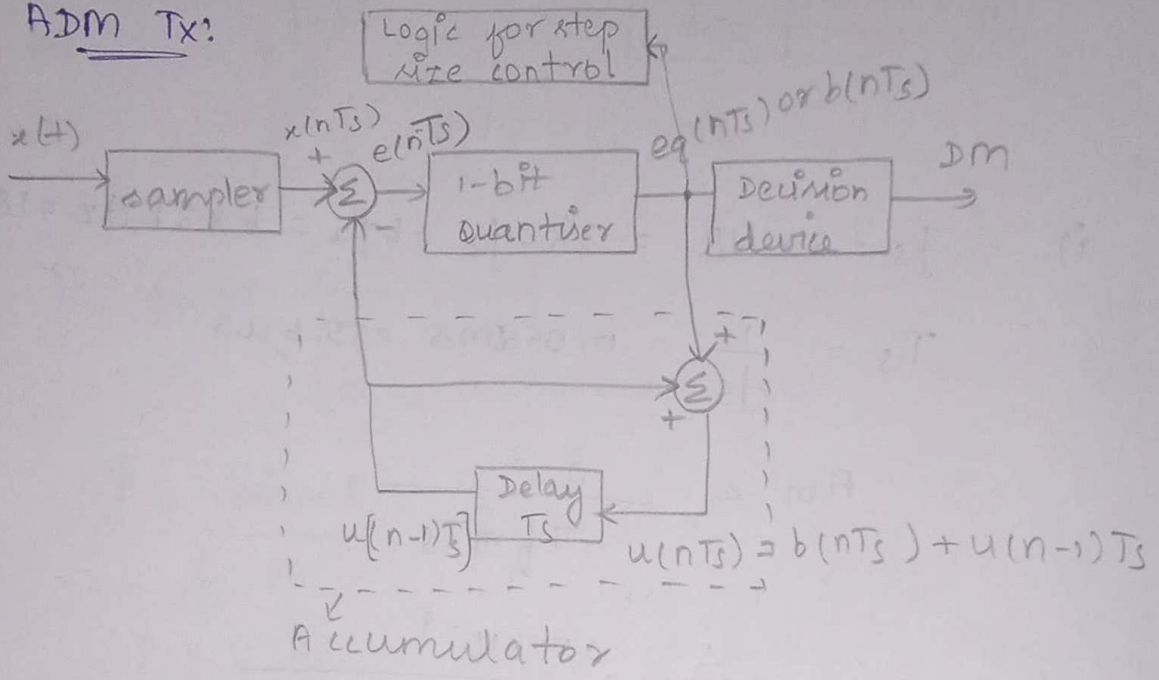
$= \frac{3}{8\pi^2 (1\times10^3)^2 (55.6\times10^{-6})^3 (3\times10^3)}$

$SNR = 43.76 \simeq 44$

ADM- Adaptive Delta Modulator:



ADM Tx:



⇒ The output of 1-bit quantiser is given to the logic for step size control. — explain in block diagram

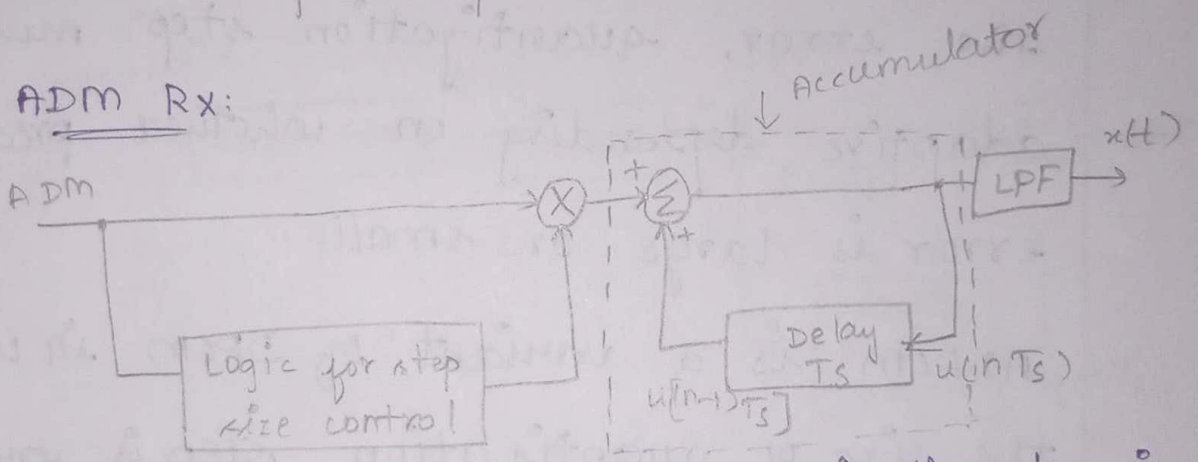
To overcome the quantisation error due to slope overload and granular noise, the step size in ADM is made adaptive to the variation in the input signal $x(t)$. In the steep segment of input signal $x(t)$, the step size is increased while when the input is varying slowly, the step size is reduced. ADM can take continuous changes or

discrete changes in step size. In Tx section, the logic for step size control is added. The step size is increased or decreased accordig to certain rules depending on 1-bit quantiser output.

For eg, if 1-bit quantiser o/p is 1 consecutively, then step size may be doubled for next sample. If quantiser o/p is 0 consecutively, then step size may be decreased.

Block diagram expln same as DM.

ADM Rx:



In ADM Rx, the first part generates the step size from each incoming bit. Exactly the same process is followed for step size control as there that in Tx. The previous i/p and present i/p decides the step size.

Then the output is given to accumulator which will generate staircase approximated waveform. Then it is fed to the LPF (smoothening filter) to get the required signal x(t).

ADPCM - Adaptive Differential Pulse Code Modulation:

=> used in speech coding

=> quantizer is non-uniform quantiser used as adaptive quantizer.

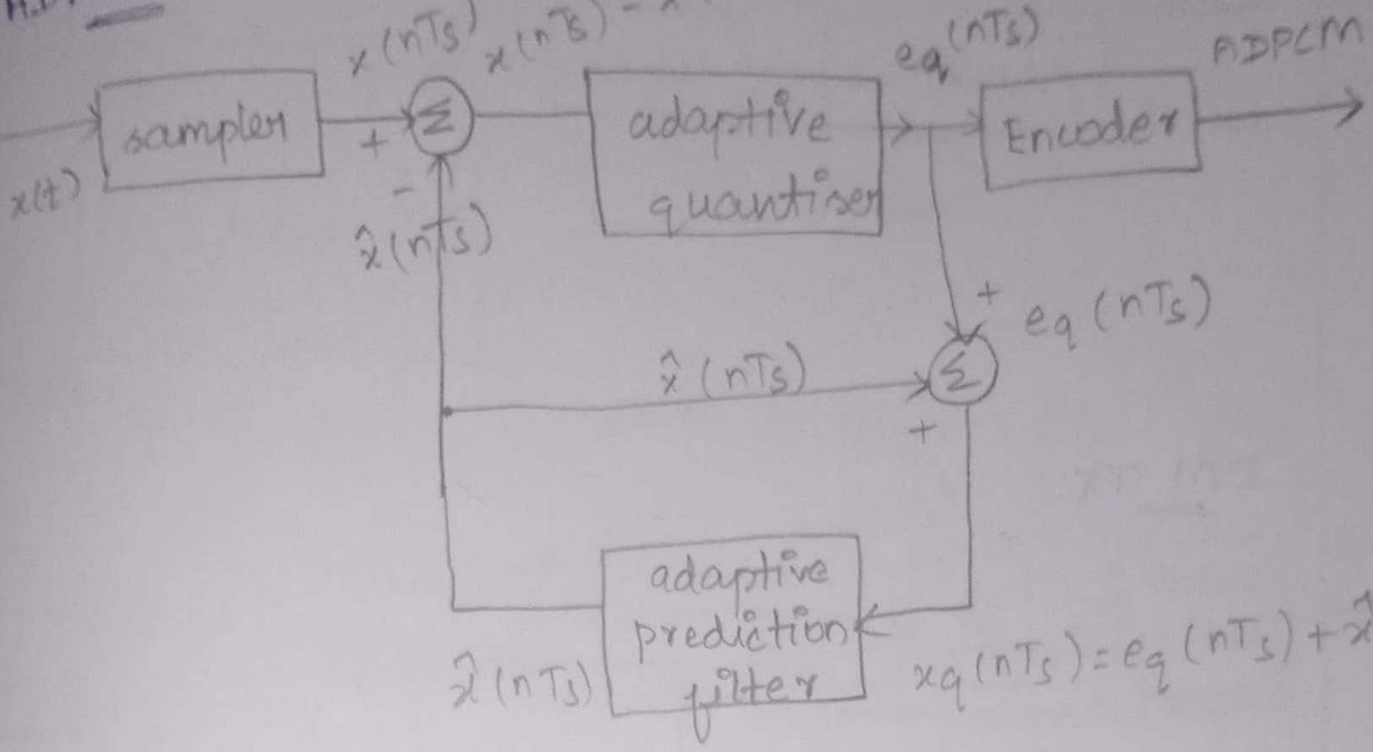
Need for ADPCM:

No. of quantization levels q in DPCM is fixed. Quantisation error occurs due to the fixed quantization step. To avoid the error, quantization step must be adaptive depending on whether prediction error is large or small.

ADPCM is a variant of DPCM in which the size of quantization step is varied to allow further reduction of the required data bandwidth for the given SNR. ADPCM improves the efficiency of DPCM by incorporating adaptive quantizer at transmitter. ADPCM also uses adaptive predictor which adjusts according to the value of each input sample and thereby reducing the no. of bits required to represent the

data sample.

ADPCM - Tx:



ADPCM - Rx:

